

In [50]:

```
#Day4
#List in python
list1 = [0,1,2,3,4]
list2 = [6,7,8,9,10]
list3 = ["Rohan",345,"Sohan", 45.67]
print("List 1 = ",list1)
print("List 2 = ",list2)
print("List 3 = ",list3)
```

```
List 1 = [0, 1, 2, 3, 4]
List 2 = [6, 7, 8, 9, 10]
List 3 = ['Rohan', 345, 'Sohan', 45.67]
```

In [51]:

```
list4 = [4,5,2,3,1]
print(list1 == list4)
list5 = [1,2,3,4,5]
print(list1 == list5)
```

```
False
False
```

In [61]:

```
#indexing and splitting the list
print(list1[0])

for i in range(0,len(list1)):
    print(list1[i], end = " ")

#splitting
print(list1[:])
print(list1[0:3])
print(list1[3:])

print(list1[-1])
print(list1[-4:-1])
print(list1[-4:])

print(list1[0:len(list1)])
print(list1[0:len(list1):2])
print(list1[0:len(list1):3]) #List_variable(start:stop:step)
```

```
0
0 1 2 3 4 [0, 1, 2, 3, 4]
[0, 1, 2]
[3, 4]
4
[1, 2, 3]
[1, 2, 3, 4]
[0, 1, 2, 3, 4]
[0, 2, 4]
[0, 3]
```

In [67]:

```
#mutable list
print(list1)
list1[3] = 567
print(list1)

list1[2:4] = [798,54] # update the 2nd and 3rd index value
print(list1)

list1[-1] = 4567
print(list1)
```

```
[0, 1, 798, 54, 33, 4]
[0, 1, 798, 567, 33, 4]
[0, 1, 798, 54, 33, 4]
[0, 1, 798, 54, 33, 4567]
```

In [75]:

```
list1 = [1,2,3,45,6]
print(list1)
list1 = list1 *2
print(list1)
list1.remove(45)
print(list1)
```

```
[1, 2, 3, 45, 6]
[1, 2, 3, 45, 6, 1, 2, 3, 45, 6]
[1, 2, 3, 6, 1, 2, 3, 45, 6]
```

In [113]:

```
list5 = list1 + list2
print(list5)
```

```
[1, 2, 3, 6, 1, 2, 3, 45, 6, 6, 7, 8, 9, 10]
```

In [114]:

```
if 200 in list5:
    print("yes")
else:
    print("No")

print(200 in list5)
```

No
False

In [115]:

```
for element in list5:
    print(element, end = " ")
```

```
1 2 3 6 1 2 3 45 6 6 7 8 9 10
```

In [116]:

```
list6 = []
print("Length = " , len(list6))
n = int(input("Enter the number"))
for i in range(0,n):
    list6.append(input("Enter the element:: "))
print("Length = " , len(list6))
print("List = ")
for var in list6:
    print(var, end = " ")
```

```
Length = 0
Enter the number3
Enter the element:: 4
Enter the element:: 67
Enter the element:: 54
Length = 3
List =
4 67 54
```

In [117]:

```
print(list5)
print("Max = ", max(list5))
print("Min = ", min(list5))
```

```
[1, 2, 3, 6, 1, 2, 3, 45, 6, 6, 7, 8, 9, 10]
Max = 45
Min = 1
```

In [118]:

```
str = "Sanjay"
list7 = list(str)
print(type(list7))
print(list7)
print(type(str))
print(str)
```

```
<class 'list'>
['S', 'a', 'n', 'j', 'a', 'y']
<class 'str'>
Sanjay
```

In [119]:

```
print(list5)
```

```
[1, 2, 3, 6, 1, 2, 3, 45, 6, 6, 7, 8, 9, 10]
```

In [120]:

```
list6 = []
for num in list5:
    if num not in list6:
        list6.append(num)
print(list6)
```

```
[1, 2, 3, 6, 45, 7, 8, 9, 10]
```

In [121]:

```
list6.sort()
print(list6)
```

[1, 2, 3, 6, 7, 8, 9, 10, 45]

In [122]:

```
list6.sort(reverse = True)
print(list6)
```

[45, 10, 9, 8, 7, 6, 3, 2, 1]

In [123]:

```
list7 = [1,2,3,4]
list8 = [3,4,5,6]
print("Common Elements: ")
for num in list7:
    for var in list8:
        if num == var:
            print(num, end = " ")
```

Common Elements:

3 4

In [124]:

```
print(list5)
num = 1
while num in list5:
    list5.remove(num)
print(list5)
```

[1, 2, 3, 6, 1, 2, 3, 45, 6, 6, 7, 8, 9, 10]
[2, 3, 6, 2, 3, 45, 6, 6, 7, 8, 9, 10]

In [125]:

```
print(len(list5))
```

12

In [126]:

```
del list5[2:4]
print(list5)
```

[2, 3, 3, 45, 6, 6, 7, 8, 9, 10]

In [127]:

```
del list5
```

In [128]:

```
print(list5)
```

-
NameError Traceback (most recent call last)
t)

<ipython-input-128-fff1b7e3f438> in <module>

----> 1 print(list5)

NameError: name 'list5' is not defined

In [131]:

```
names = ["Fab","Tom","Joe","Schidlt","Pyhton"]  
print(names)  
names.sort()  
print(names)
```

```
['Fab', 'Tom', 'Joe', 'Schidlt', 'Pyhton']  
['Fab', 'Joe', 'Pyhton', 'Schidlt', 'Tom']
```

In [132]:

```
num1 = [1,2,3,4,5]  
num2 = [3,5,7,8,0]  
print(list(set(num1) - set(num2)))
```

```
[1, 2, 4]
```

In [133]:

```
print(list(set(num2) - set(num1)))
```

```
[0, 8, 7]
```

In [142]:

```
print(list6)  
print("Index of 8 = ",list6.index(8))  
print("Index of Max element = ",list6.index(max(list6)))  
print("Index of Min element = ",list6.index(min(list6)))
```

```
[45, 10, 9, 8, 7, 6, 3, 2, 1]  
Index of 8 = 3  
Index of Max element = 0  
Index of Min element = 8
```

In [143]:

```
def func(varlist1):
    varlist2 = []
    for num in varlist1:
        if num not in varlist2:
            varlist2.append(num)
    return varlist2

varlist1 = [34,564,"Joe",765,3,564,5,67,34,"Rohan",865,5,4,"Rohan",3]
print(varlist1)
print(func(varlist1))
```

```
[34, 564, 'Joe', 765, 3, 564, 5, 67, 34, 'Rohan', 865, 5, 4, 'Rohan', 3]
[34, 564, 'Joe', 765, 3, 5, 67, 'Rohan', 865, 4]
```

In [148]:

```
varlist3 = [45,65,33,567,8,96,5,35,32,66,68,32,56,89]

evenlist = []
oddlist = []
for num in varlist3:
    if num%2 == 0:
        evenlist.append(num)
    else:
        oddlist.append(num)
print("Even List = " , evenlist)
print("Odd List = ", oddlist)

evenlist = []
oddlist = []

for i in range(0,len(varlist3)):
    if varlist3[i]%2 == 0:
        evenlist.append(varlist3[i])
    else:
        oddlist.append(varlist3[i])
print("Even List = " , evenlist)
print("Odd List = ", oddlist)
```

```
Even List = [8, 96, 32, 66, 68, 32, 56]
Odd List = [45, 65, 33, 567, 5, 35, 89]
Even List = [8, 96, 32, 66, 68, 32, 56]
Odd List = [45, 65, 33, 567, 5, 35, 89]
```

In [150]:

```
print(varlist3)
varlist4 = []
n1 = int(input("Enter a number:: "))
n2 = int(input("Enter a number:: "))
for num in varlist3:
    if num%n1 == 0 and num%n2 == 0:
        varlist4.append(num)

print(varlist4)
```

[45, 65, 33, 567, 8, 96, 5, 35, 32, 66, 68, 32, 56, 89]

Enter a number:: 2

Enter a number:: 4

[8, 96, 32, 68, 32, 56]

In [151]:

```
print(varlist3)
square = []
cube = []

for num in varlist3:
    square.append(num ** 2)
    cube.append(num ** 3)
print(square)
print(cube)
```

[45, 65, 33, 567, 8, 96, 5, 35, 32, 66, 68, 32, 56, 89]

[2025, 4225, 1089, 321489, 64, 9216, 25, 1225, 1024, 4356, 4624, 1024, 3136, 7921]

[91125, 274625, 35937, 182284263, 512, 884736, 125, 42875, 32768, 287496, 314432, 32768, 175616, 704969]

In [155]:

```
square_cube = []
print(varlist3)
for num in varlist3:
    if num%2 == 0:
        square_cube.append(num ** 2)
    else:
        square_cube.append(num ** 3)
print(square_cube)
```

[45, 65, 33, 567, 8, 96, 5, 35, 32, 66, 68, 32, 56, 89]

[91125, 274625, 35937, 182284263, 64, 9216, 125, 42875, 1024, 4356, 4624, 1024, 3136, 704969]

In [166]:

```
print(varlist3)
def square(list1):
    square1 = []
    for num in list1:
        square1.append(num ** 2)
    return square1

def cube(list2):
    cube1 = []
    for num in list2:
        cube1.append(num ** 3)
    return cube1

square = square(varlist3)
cube = cube(varlist3)
print("Square of numbers = ", square)
print("Cube of numbers = ", cube)
```

```
[45, 65, 33, 567, 8, 96, 5, 35, 32, 66, 68, 32, 56, 89]
Square of numbers = [2025, 4225, 1089, 321489, 64, 9216, 25, 1225, 1024,
4356, 4624, 1024, 3136, 7921]
Cube of numbers = [91125, 274625, 35937, 182284263, 512, 884736, 125, 428
75, 32768, 287496, 314432, 32768, 175616, 704969]
```

In [172]:

```
Squ_cub = square[0:len(square)//2] + cube[len(cube)//2:len(cube)]
print(Squ_cub)
```

```
[2025, 4225, 1089, 321489, 64, 9216, 25, 42875, 32768, 287496, 314432, 327
68, 175616, 704969]
```

In [177]:

```
print(varlist3)
revlist3 = varlist3[::-1]
print(revlist3)
```

```
[5, 8, 32, 32, 33, 35, 45, 56, 65, 66, 68, 89, 96, 567]
[567, 96, 89, 68, 66, 65, 56, 45, 35, 33, 32, 32, 8, 5]
```

In []: